

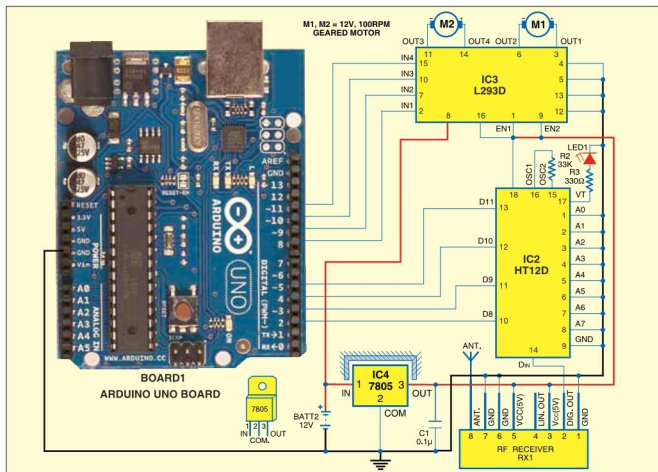


Fig. 3: Circuit of receiver and motor driver

The internal oscillation frequency of decoder HT12D is 50 times the oscillation frequency of encoder HT12E. The HT12D receives the data from the HT12E on its D_{IN} pin serially. If the address part of the data received

PARTS LIST

Semiconductors:

- IC1 - HT12E encoder
- IC2 - HT12D decoder
- IC3 - L293D motor driver
- IC4 - 7805, 5V regulator
- LED1 - 5mm LED
- TX1 - 433MHz RF ASK transmitter
- RX1 - 433MHz RF ASK receiver
- BOARD1 - Arduino UNO board

Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ per cent carbon):

- R1 - 750-kilo-ohm
- R2 - 33-kilo-ohm
- R3 - 330-ohm

Capacitors:

- C1 - 0.1 μ F ceramic disk

Miscellaneous:

- S1-S4 - Push-to-on tactile switch
- M1, M2 - 12V, 100-rpm geared motor
- BATT1 - 9V battery
- BATT2 - 12V battery

matches the levels on A0 through A7 pins four times in succession, the valid transmission (VT) pin is taken high. The data on pins AD8 through AD11 of the HT12E appears on pins D8 through D11 of the HT12D. Thus the device acts as a receiver of 4-bit data (16 possible codes) with 8-bit addressing (256 possible channels).

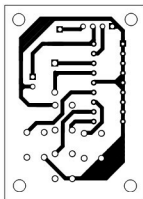


Fig. 4: An actual-size, single-side PCB for the RF transmitter

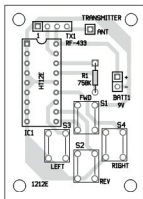


Fig. 5: Component layout for the PCB in Fig. 4

Transmitter circuit. Switches S1, S2, S3 and S4 are interfaced with AD8 through AD11 of encoder HT12E for forward (FWD), reverse (REV), left (LEFT) and right (RIGHT) motions, respectively. Resistor R1 is connected between oscillator pins 15 and 16 to set the transmitter frequency.

HT12E is permanently enabled for transmission by connecting its $\overline{TE}$ pin to ground. When any switch, say, S1, is pressed, the corresponding data is serially transmitted from D_{OUT} pin through the RF ASK transmitter module. A 9V battery is used to power the circuit.

Receiver and motor driver circuit. Assuming that address pins on the encoder and the decoder